

Compiling for a device that keeps moving
Placement and routing at 78.5 against a 283.5 baseline, and what the compiled circuit is worth at run time

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A routed circuit is a compile-time object. What a user gets is its success on a device that has drifted since it was characterized.

1. The solver

The rules decide the design. The 2Q list must be reproduced exactly, in order; only SWAPs and the placement are free; depth is the ASAP critical path. So: placement, which qubit moves along which path, and SWAPs that ride in idle slots.

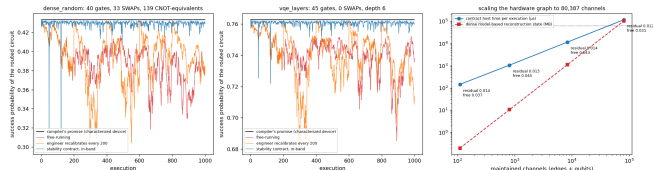
Placement: interaction graph weighted by gate multiplicity (earlier gates count more), simulated annealing over injective placements.

Routing: gates in order; candidate SWAPs must bring the current gate one step closer (minimum $d-1$ SWAPs, no oscillation); chosen by SABRE lookahead over the next 8 gates plus a charge for the ASAP layer the SWAP lands on.

Refinement: three SABRE reverse passes, 120 restarts over three seeds, exact scorer in the loop, deterministic.

benchmark	baseline	ours	SWAPs	depth
ghz_star	14.0	7.0	3	8
chain_trotter	15.0	4.5	0	9
ladder_trotter	35.5	6.5	3	7
qaoa_random	39.0	11.5	6	11
dense_random	122.0	46.0	33	26
vqe_layers	58.0	3.0	0	6 (optimal)
total	283.5	78.5		

2. What the circuit is worth at run time



Every edge and qubit drifts after characterization: 4% wander plus a slow walk. Gate infidelity = floor + $\varepsilon^2/4$; a SWAP is three CNOTs.

dense_random, 1,000 executions: promise 0.433. Free-running ends at 0.373. Engineer every 200: 0.390, dips to 0.304. Contract: **0.429**, 99% of the promise, 81 μ s per execution.

vqe_layers: 0.763 promised; contract 0.760.

The morning re-map exists because the compiler's cost model goes stale. Maintenance keeps it true.

3. Why this is the only one that scales

qubits	channels	contract host / execution	residual (free)	dense reconstruction model
48	111	146 μ s	0.013 (0.037)	0.2 MB
336	815	1.1 ms	0.013 (0.045)	11 MB
3,360	8,319	11.6 ms	0.014 (0.043)	1.1 GB
32,256	80,387	118 ms	0.012 (0.031)	103 GB, beyond a 64 GB host

Three ways to keep a device where the compiler characterized it. An engineer on standby at every site cannot follow a wander faster than the schedule. Reconstructing the device’s state costs n^2 memory and n^3 setup, and stops at a few thousand channels. An in-band contract is linear in channels and never touches the state.

Honest limits: a simulated device with measured drift structure, stated floors; the probe rung must be an odd multiple of $\pi/2$ with bounds inside its monotone range, which we learned by breaking it; no Stretch Goal A. The experiment runs as written on hardware: probes are circuits, corrections are angle scales.